익명 네트워크: 프라이버시를 위한 P2P

1. 들어가며

지금까지 우리는 P2P의 진화를 살펴보았습니다. Napster에서 시작해 DHT, BitTorrent, 블록체인까지—이 시스템들은 모두 **효율성**과 **분산화**에 집중했습니다. 하지만 누가 누구와 통신하는지, 어떤 콘텐츠에 접근하는지는 여전히 노출되었습니다.

2004년, Roger Dingledine, Nick Mathewson, Paul Syverson은 이 문제를 해결하기 위한 시스템을 발표했습니다:

> "We present Tor, a circuit-based low-latency anonymous communication service. This second-generation Onion Routing system addresses limitations in the original design by adding perfect forward secrecy, congestion control, directory servers, integrity checking, configurable exit policies, and a practical design for location-hidden services via rendezvous points."
> — Dingledine et al., "Tor: The Second-Generation Onion Router", USENIX Security 2004

이 글에서는 Tor의 원본 논문과 I2P 기술 문서를 통해 익명 네트워크의 설계를 탐구합니다.

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2. 왜 익명성이 필요한가

2.1 프라이버시의 중요성

Tor 논문은 익명 통신의 필요성을 명확히 합니다:

> "Onion Routing is a distributed overlay network designed to anonymize TCP-based applications like web browsing, secure shell, and instant messaging. Clients choose a path through the network and build a circuit, in which each node (or 'onion router' or 'OR') in the path knows its predecessor and successor, but no other nodes in the circuit."
> — Dingledine et al., 2004, Section 1

2.2 설계 목표

Tor의 핵심 목표들:

> "**Deployability**: The design must be deployed and used in the real world. Thus it must not be expensive to run; must not place a heavy liability burden on operators; and must not be difficult or expensive to implement."
> — Dingledine et al., 2004, Section 3

> "**Usability**: A hard-to-use system has fewer users—and because anonymity systems hide users among users, a system with fewer users provides less anonymity. Usability is thus not only a convenience: it is a security requirement."
> — Dingledine et al., 2004, Section 3

핵심 통찰:

> **"익명 시스템에서 사용성은 보안의 일부다. 사용자가 적으면 숨을 군중도 작아진다."**

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3. Onion Routing의 원리

3.1 기본 개념

Onion Routing의 핵심 아이디어:

> "Traffic flows down the circuit in fixed-size cells, which are unwrapped by a symmetric key at each node (like the layers of an onion) and relayed downstream."
> — Dingledine et al., 2004, Section 1

```
┌─────────┐     ┌─────────┐     ┌─────────┐     ┌─────────┐
│  Alice  │ ──→ │   OR1   │ ──→ │   OR2   │ ──→ │   OR3   │ ──→ Bob
└─────────┘     └─────────┘     └─────────┘     └─────────┘
                  (Guard)       (Middle)         (Exit)

암호화 레이어:  [[[메시지]K3]K2]K1
                      ↓
OR1이 K1으로 복호화: [[메시지]K3]K2
                      ↓
OR2가 K2로 복호화:   [메시지]K3
                      ↓
OR3가 K3로 복호화:    메시지 → Bob
```

3.2 각 노드가 아는 것

> "Each node... knows its predecessor and successor, but no other nodes in the circuit."
> — Dingledine et al., 2004, Section 1

노드	알 수 있는 것	알 수 없는 것
OR1 (Guard)	Alice의 IP, OR2	최종 목적지, 메시지 내용
OR2 (Middle)	OR1, OR3	Alice, 최종 목적지, 메시지
OR3 (Exit)	OR2, Bob	Alice, 전체 경로

3.3 고정 크기 셀

> "Traffic passes along these connections in fixed-size cells. Each cell is 512 bytes, and consists of a header and a payload."
> — Dingledine et al., 2004, Section 4.1

고정 크기는 트래픽 분석을 어렵게 만듭니다.

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4. Tor의 핵심 혁신

4.1 Perfect Forward Secrecy

1세대 Onion Routing의 문제:

> "Onion Routing was originally vulnerable to a single hostile node recording traffic and later compromising successive nodes in the circuit and forcing them to decrypt it."
> — Dingledine et al., 2004, Section 1

Tor의 해결책:

> "Rather than using a single multiply encrypted data structure (an onion) to lay each circuit, Tor now uses an incremental or telescoping path-building design, where the initiator negotiates session keys with each successive hop in the circuit. Once these keys are deleted, subsequently compromised nodes cannot decrypt old traffic."
> — Dingledine et al., 2004, Section 1

4.2 회로 구축 (Telescoping)

> "A user's OP constructs circuits incrementally, negotiating a symmetric key with each OR on the circuit, one hop at a time."
> — Dingledine et al., 2004, Section 4.2

단계별 과정:

1. Alice → OR1: Diffie-Hellman 협상
2. Alice → OR1 → OR2: OR1을 통해 OR2와 DH 협상
3. Alice → OR1 → OR2 → OR3: OR1, OR2를 통해 OR3와 DH 협상

> "As a side benefit, onion replay detection is no longer necessary, and the process of building circuits is more reliable, since the initiator knows when a hop fails and can then try extending to a new node."
> — Dingledine et al., 2004, Section 1

4.3 스트림 다중화

> "Onion Routing originally built a separate circuit for each TCP stream. Because building a circuit can take several tenths of a second (due to public-key cryptography and network latency), this design imposed high costs on applications like web browsing that open many TCP streams."
> — Dingledine et al., 2004, Section 4.2

Tor의 해결책:

> "In Tor, each circuit can be shared by many TCP streams... OPs consider making a new circuit once a minute: thus even heavy users spend negligible time building circuits, but a limited number of requests can be linked to each other through a given exit node."
> — Dingledine et al., 2004, Section 4.2

4.4 Leaky-Pipe Topology

흥미로운 기능:

> "Through in-band signaling within the circuit, Tor initiators can direct traffic to nodes partway down the circuit. This novel approach allows traffic to exit the circuit from the middle—possibly frustrating traffic shape and volume attacks based on observing the end of the circuit."
> — Dingledine et al., 2004, Section 1

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5. 위협 모델과 한계

5.1 Tor가 방어하는 것

> "We assume an adversary who can observe some fraction of network traffic; who can generate, modify, delete, or delay traffic; who can operate onion routers of his own; and who can compromise some fraction of the onion routers."
> — Dingledine et al., 2004, Section 3.1

5.2 Tor가 방어하지 못하는 것

Global Passive Adversary에 대한 한계:

> "Like all practical low-latency systems, Tor does not protect against such a strong adversary."
> — Dingledine et al., 2004, Section 3.1

End-to-End 타이밍 공격:

> "By observing both ends, passive attackers can confirm a suspicion that Alice is talking to Bob if the timing and volume patterns of the traffic on the connection are distinct enough; active attackers can induce timing signatures on the traffic to force distinct patterns."
> — Dingledine et al., 2004, Section 3.1

Tor의 현실적인 목표:

> "Rather than focusing on these traffic confirmation attacks, we aim to prevent traffic analysis attacks, where the adversary uses traffic patterns to learn which points in the network he should attack."
> — Dingledine et al., 2004, Section 3.1

5.3 명시적 비목표

> "**Not peer-to-peer**: Tarzan and MorphMix aim to scale to completely decentralized peer-to-peer environments with thousands of short-lived servers, many of which may be controlled by an adversary. This approach is appealing, but still has many open problems."
> — Dingledine et al., 2004, Section 3

> "**Not secure against end-to-end attacks**: Tor does not claim to provide a definitive solution to end-to-end timing or intersection attacks."
> — Dingledine et al., 2004, Section 3

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6. Directory Servers

6.1 네트워크 상태 관리

> "First-generation Onion Routing designs used in-band network status updates: each router flooded a signed statement to its neighbors, which propagated it onward. But anonymizing networks have different security goals than typical link-state routing protocols."
> — Dingledine et al., 2004, Section 5.3

6.2 중앙화의 이유

> "Tor uses a small group of redundant, well-known onion routers to track changes in network topology and node state, including keys and exit policies. Each such directory server acts as an HTTP server, so clients can fetch current network state and router lists."
> — Dingledine et al., 2004, Section 5.3

6.3 Sybil 공격 방어

> "When a directory server receives a signed statement for an OR, it checks whether the OR's identity key is recognized. Directory servers do not automatically advertise unrecognized ORs. (If they did, an adversary could take over the network by creating many servers.)"
> — Dingledine et al., 2004, Section 5.3

---

7. Hidden Services (Rendezvous Points)

7.1 양방향 익명성

> "Rendezvous points are a building block for location-hidden services (also known as responder anonymity) in the Tor network. Location-hidden services allow Bob to offer a TCP service, such as a webserver, without revealing his IP address."
> — Dingledine et al., 2004, Section 4.7

7.2 설계 목표

> "**Access-controlled**: Bob needs a way to filter incoming requests, so an attacker cannot flood Bob simply by making many connections to him."
> — Dingledine et al., 2004, Section 4.7

> "**Robust**: Bob should be able to maintain a long-term pseudonymous identity even in the presence of router failure. Bob's service must not be tied to a single OR."
> — Dingledine et al., 2004, Section 4.7

> "**Smear-resistant**: A social attacker who offers an illegal or disreputable location-hidden service should not be able to 'frame' a rendezvous router by making observers believe the router created that service."
> — Dingledine et al., 2004, Section 4.7

7.3 Rendezvous 프로토콜

논문의 Appendix A에서 설명하는 단계:

1. Bob이 introduction points를 선택하고 DHT에 광고
2. Bob이 각 introduction point로 회로 구축
3. Alice가 Bob의 서비스 정보를 DHT에서 조회
4. Alice가 rendezvous point(RP)를 선택하고 회로 구축
5. Alice가 Bob의 introduction point를 통해 RP 정보 전달
6. Bob이 RP로 회로 구축하고 연결 완료

```
Alice                    Introduction Point              Bob
  |                            |                          |
  |----[RP 정보, DH g^x]----→ |                          |
  |                            |----[RP 정보, DH g^x]---→|
  |                            |                          |
  |                                                       |
  |              Rendezvous Point                         |
  |                    |                                  |
  |←--[회로 구축]-----|←--------[회로 구축, DH g^y]-------|
  |                    |                                  |
  |←============ 익명 연결 완료 ==========→|
```

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8. I2P: 다른 접근법

8.1 Invisible Internet Project

I2P는 Tor와 다른 설계 철학을 가진 익명 네트워크입니다:

> "I2P (Invisible Internet Project) is a decentralized message oriented mixnet, that allows users to communicate in an anonymous and encrypted manner."
> — I2P Technical Documentation

8.2 Garlic Routing

I2P의 핵심 차별점은 **Garlic Routing**입니다:

> "Garlic routing is a variant of onion routing that encrypts multiple messages together to make it more difficult for attackers to perform traffic analysis while also increasing the speed of data transfer."
> — Wikipedia, Garlic Routing

I2P 공식 문서에서:

> "Michael J. Freedman defined garlic routing as an extension of onion routing, in which multiple messages are bundled together. He called each message a 'bulb', whereas I2P calls them 'garlic cloves.'"
> — I2P Documentation, Garlic Routing

8.3 Onion vs Garlic

특성	Onion Routing (Tor)	Garlic Routing (I2P)
메시지	단일 메시지	여러 메시지 번들
터널 방향	양방향 회로	단방향 터널
목적	Outproxy (인터넷 접근)	내부 서비스 (Eepsites)
라우팅	회로 스위칭	패킷 스위칭

8.4 단방향 터널

I2P의 핵심 설계:

> "In I2P, tunnels are unidirectional. Each party builds two tunnels, one for outbound and one for inbound traffic. Therefore, four tunnels are required for a single round-trip message and reply."
> — I2P Documentation, Garlic Routing

```
Alice의 통신:

[Alice] ──Outbound Tunnel──→ [Gateway] ──→ [Bob's Inbound Gateway] ──Inbound Tunnel──→ [Bob]

[Alice] ←──Inbound Tunnel──← [Gateway] ←── [Bob's Outbound Gateway] ←──Outbound Tunnel── [Bob]
```

8.5 네트워크 데이터베이스

I2P는 Kademlia 기반 DHT를 사용합니다:

> "Each client application has their own I2P 'router' that finds other clients by querying against the fully distributed 'network database' - a custom structured distributed hash table (DHT) based on the Kademlia algorithm."
> — Whonix I2P Documentation

RouterInfo: 라우터의 기능과 전송 주소
LeaseSet: 목적지의 터널과 암호화 키

---

9. Tor vs I2P 비교

9.1 설계 철학

측면	Tor	I2P
주요 용도	인터넷 익명 접근	내부 익명 서비스
네트워크 구조	중앙 Directory	분산 DHT
터널 구조	양방향 회로	단방향 터널
Exit 노드	핵심 기능	선택적 (Outproxy)
Hidden Services	.onion	.i2p (Eepsites)

9.2 I2P의 장점

I2P 문서에서 언급하는 강점들:

> "In comparison, I2P's primary strengths are: optimization for hidden services; a fully distributed design; better peer selection; varied and untrusted directory servers; peer-to-peer friendly nature; improved load balancing and resilience; unidirectional tunnels; protection against client activity detection; short-

> lived tunnels; low bandwidth overhead for full peers."
> — Whonix I2P Documentation

9.3 각 시스템의 적합한 용도

Tor가 더 적합한 경우:

- 일반 웹 브라우징
- 외부 인터넷 서비스 접근
- 일회성 익명 접속

I2P가 더 적합한 경우:

- 익명 P2P 파일 공유
- 장기 익명 서비스 운영
- 내부 커뮤니티 서비스

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10. 공격과 방어

10.1 Tor가 방어하는 공격들

Passive Attacks:

> "Observing a user's connection will not reveal her destination or data, but it will reveal traffic patterns (both sent and received)."
> — Dingledine et al., 2004, Section 6

Active Attacks - Key Compromise:

> "An attacker who learns the TLS session key can see control cells and encrypted relay cells on every circuit on that connection; learning a circuit session key lets him unwrap one layer of the encryption."
> — Dingledine et al., 2004, Section 6

키 순환의 중요성:

> "Periodic key rotation limits the window of opportunity for these attacks."
> — Dingledine et al., 2004, Section 6

10.2 어려운 공격들

End-to-End Correlation:

> "Tor only minimally hides such correlations. An attacker watching patterns of traffic at the initiator and the responder will be able to confirm the correspondence with high probability."
> — Dingledine et al., 2004, Section 6

Website Fingerprinting:

> "Rather than searching exit connections for timing and volume correlations, the adversary may build up a database of 'fingerprints' containing file sizes and access patterns for targeted websites."
> — Dingledine et al., 2004, Section 6

10.3 Directory 공격

> "Subvert a majority of directory servers: An adversary who controls more than half the directory servers can include as many compromised ORs in the final directory as he wishes. We must ensure that directory server operators are independent and attack-resistant."
> — Dingledine et al., 2004, Section 6

---

11. 혼잡 제어

11.1 문제 정의

> "Even with bandwidth rate limiting, we still need to worry about congestion, either accidental or intentional. If enough users choose the same OR-to-OR connection for their circuits, that connection can become saturated."
> — Dingledine et al., 2004, Section 4.6

11.2 회로 레벨 조절

> "To control a circuit's bandwidth usage, each OR keeps track of two windows. The packaging window tracks how many relay data cells the OR is allowed to package for transmission back to the OP, and the delivery window tracks how many relay data cells it is willing to deliver to TCP streams outside the network."
> — Dingledine et al., 2004, Section 4.6

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12. Exit Policy와 남용 방지

12.1 자원봉사자 보호

> "Exit abuse is a serious barrier to wide-scale Tor deployment. Anonymity presents would-be vandals and abusers with an opportunity to hide the origins of their activities."
> — Dingledine et al., 2004, Section 5.2

12.2 Exit Policy

> "To mitigate abuse issues, in Tor, each onion router's exit policy describes to which external addresses and ports the router will connect."
> — Dingledine et al., 2004, Section 5.2

유형들:

- **Open exit**: 어디든 연결
- **Restricted exit**: 특정 포트/서비스만 허용
- **Middleman**: 다른 Tor 노드로만 릴레이
- **Private exit**: 로컬 호스트/네트워크만 연결

12.3 현실적 고려

> "Finally, we note that exit abuse must not be dismissed as a peripheral issue: when a system's public image suffers, it can reduce the number and diversity of that system's users, and thereby reduce the anonymity of the system itself. Like usability, public perception is a security parameter."
> — Dingledine et al., 2004, Section 5.2

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13. P2P와 익명성의 교차점

13.1 P2P 기술의 활용

익명 네트워크는 이전 편들에서 다룬 P2P 기술들을 적극 활용합니다:

기술	익명 네트워크에서의 활용
DHT (2편)	I2P 네트워크 데이터베이스
분산 검색	Hidden service 발견
Overlay Network	전체 아키텍처
인센티브 (3편)	대역폭 기여 (I2P)

13.2 차이점

하지만 익명 네트워크는 추가적인 요구사항이 있습니다:

1. **메타데이터 보호**: 누가 누구와 통신하는지 숨김
2. **트래픽 분석 저항**: 패턴으로 추적 불가
3. **Forward Secrecy**: 과거 통신 보호

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14. 열린 문제들

Tor 논문은 여전히 해결되지 않은 문제들을 나열합니다:

14.1 경로 길이

> "How should we choose path lengths? If Alice only ever uses two hops, then both ORs can be certain that by colluding they will learn about Alice and Bob."
> — Dingledine et al., 2004, Section 8

14.2 회로 교체 빈도

> "How often should users rotate to fresh circuits? Frequent rotation is inefficient, expensive, and may lead to intersection attacks and predecessor attacks, but infrequent rotation makes the user's traffic linkable."
> — Dingledine et al., 2004, Section 8

14.3 확장성

> "To scale to many users, and to prevent an attacker from observing the whole network, it may be necessary to support far more servers than Tor currently anticipates."

> — Dingledine et al., 2004, Section 8

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15. 시리즈 마무리: P2P의 여정

15.1 진화의 궤적

```
1999: Napster     - P2P의 시작, 중앙 인덱스
        ↓
2000: Gnutella    - 완전 분산, Flooding
        ↓
2001: Chord/DHT   - 구조화된 O(log N) 검색
        ↓
2001: BitTorrent  - 인센티브 설계
        ↓
2004: Tor         - 익명 통신
        ↓
2008: Bitcoin     - 분산 합의
        ↓
2014: Ethereum    - 프로그래머블 블록체인
        ↓
현재: libp2p      - 모듈형 P2P 스택
```

15.2 각 세대가 해결한 문제

세대	시스템	해결한 문제	핵심 기술
1세대	Napster	파일 공유	중앙 인덱스
1세대	Gnutella	분산 검색	Flooding
2세대	Chord	확장성	DHT
2세대	Kademlia	효율성	XOR 거리
3세대	BitTorrent	Free Riding	Tit-for-Tat
4세대	Tor	프라이버시	Onion Routing
4세대	I2P	내부 익명 서비스	Garlic Routing
5세대	Bitcoin	분산 합의	PoW
5세대	Ethereum	스마트 컨트랙트	EVM

15.3 핵심 교훈

1. **분산화의 스펙트럼**: 완전 중앙화(Napster)에서 완전 분산(Gnutella)까지, 각각의 트레이드오프가 있다
2. **구조의 힘**: DHT는 무구조(Gnutella)와 중앙화(Napster) 사이의 균형을 찾았다
3. **인센티브의 중요성**: BitTorrent는 이기심을 협력으로 바꾸는 방법을 보여주었다

4. **프라이버시는 기능이다**: Tor는 "누가 누구와 통신하는가"를 숨기는 것도 분산 시스템의 과제임을 보여주었다
5. **합의는 어렵다**: 블록체인은 신뢰 없는 합의가 가능하지만 비용이 크다

---

16. 마치며

Tor 논문의 결론:

> "Tor works on the real-world Internet, requires no special privileges or kernel modifications, requires little synchronization or coordination between nodes, and provides a reasonable tradeoff between anonymity, usability, and efficiency."
> — Dingledine et al., 2004, Abstract

그리고 중요한 철학:

> "Usability is thus not only a convenience: it is a security requirement."
> — Dingledine et al., 2004, Section 3

P2P 시리즈의 핵심 메시지

> **"P2P는 단순한 기술이 아니라 철학이다. 중앙 권위 없이 협력하고, 신뢰 없이 합의하며, 감시 없이 통신하는 것—이것이 P2P가 추구하는 미래다."**

참고 자료

논문 (직접 인용)

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   - Tor의 창시자들이 쓴 원본 논문

기술 문서

2. **I2P Documentation** - Garlic Routing
   - https://geti2p.net/en/docs/how/garlic-routing
3. **I2P Technical Introduction**
   - https://geti2p.net/en/docs/how/tech-intro
4. **Wikipedia** - Garlic Routing
   - https://en.wikipedia.org/wiki/Garlic_routing

시리즈 이전 편

- **1편**: 초기 P2P (Napster, Gnutella, FastTrack)
- **2편**: DHT (Chord, Kademlia)

- **3편**: BitTorrent (인센티브 설계)
- **4편**: 블록체인 P2P (Bitcoin, Ethereum, libp2p)